Executive Summary

SunRise Renewables Pvt. Ltd.

1 GW Utility-Scale Solar PV Project Financial Model

1. Project Overview

Metric	Value
Installed Capacity	**1,000 MW**
Installed DC Capacity	**1,350 MWp**
Construction Period	**24 Months**
COD	**01-Apr-28**
PPA Tenure	**25 Years**
Model Period	**FY27 – FY53**

2. Funding Summary

Metric	Value
Total Project Cost	**₹4,741.81 Cr**
Debt Funding	**₹3,319.27 Cr**
Equity Funding	**₹1,422.54 Cr**
Debt to Equity	**70%:30%**
Interest Rate	**9.25%**
Loan Tenure	**18 Years**
Moratorium	**3 Years**

3. Operational Assumptions

Metric	Value
PPA Tariff	**₹3.00/kWh**
Plant Load Factor	**24%**
Module Degradation	**0.50% p.a.**
Plant Availability	**99%**
Grid Availability	**98.50%**
Auxiliary Consumption	**1.20%**
Collection Period	**60 Days**

4. Investment Evaluation

Metric	Value
WACC	**10.98%**
Project IRR	**6.82%**
Equity IRR	**7.46%**
Project NPV	**₹(1,113.38) Cr**
Equity NPV	**₹(613.33) Cr**
Project Payback	**12.80 Years**
Equity Payback	**18.97 Years**

5. Debt Service Metrics

Metric	Value
Minimum DSCR	**0.79×**
Average DSCR	**1.10×**
LLCR	**1.12×**

6. Sensitivity Analysis

in ₹ crores		Tariff per unit				
		2	2.5	3	3.5	4
PLF	22%	(2,447.02)	(1,998.71)	(1,550.41)	(1,102.10)	(653.79)
	23%	(2,365.51)	(1,896.83)	(1,428.14)	(959.46)	(490.77)
	24%	(2,284.00)	(1,794.94)	(1,305.88)	(816.81)	(327.75)
	25%	(2,202.49)	(1,693.05)	(1,183.61)	(674.17)	(164.73)
	26%	(2,120.98)	(1,591.16)	(1,061.35)	(531.53)	(1.71)

in ₹ crores		O&M Escalation				
		3%	4%	5%	6%	7%
Interest rate	8.50%	(1,206.79)	(1,253.89)	(1,307.26)	(1,367.87)	(1,436.85)
	9.00%	(1,205.87)	(1,252.97)	(1,306.34)	(1,366.95)	(1,435.92)
	9.25%	(1,205.41)	(1,252.51)	(1,305.88)	(1,366.48)	(1,435.46)
	10.00%	(1,204.02)	(1,251.12)	(1,304.49)	(1,365.10)	(1,434.07)
	10.50%	(1,203.10)	(1,250.20)	(1,303.57)	(1,364.17)	(1,433.15)

6. Investment Decision

Criteria	Status
Project NPV > 0	✗ Not Met
Project IRR > WACC	✗ Not Met
Equity IRR > WACC	✗ Not Met
Average DSCR > 1.35	✗ Not Met
LLCR > 1.40	✗ Not Met
Payback < 12 Years	✗ Not Met

7. Key Takeaways

Utility-scale **1 GW** solar PV project developed under a **25-year PPA**.

Total project investment of **₹4,741.81 Cr**, financed through a **70:30 debt-equity structure**.

The current base-case assumptions result in a **negative NPV**, **IRRs below the cost of capital**, and **debt service metrics below lender thresholds**.

Based on the current assumptions, the project is **not financially viable** and would require improvements such as a higher tariff, lower capital cost, cheaper financing, enhanced PLF, or revised operating assumptions to meet investment criteria.